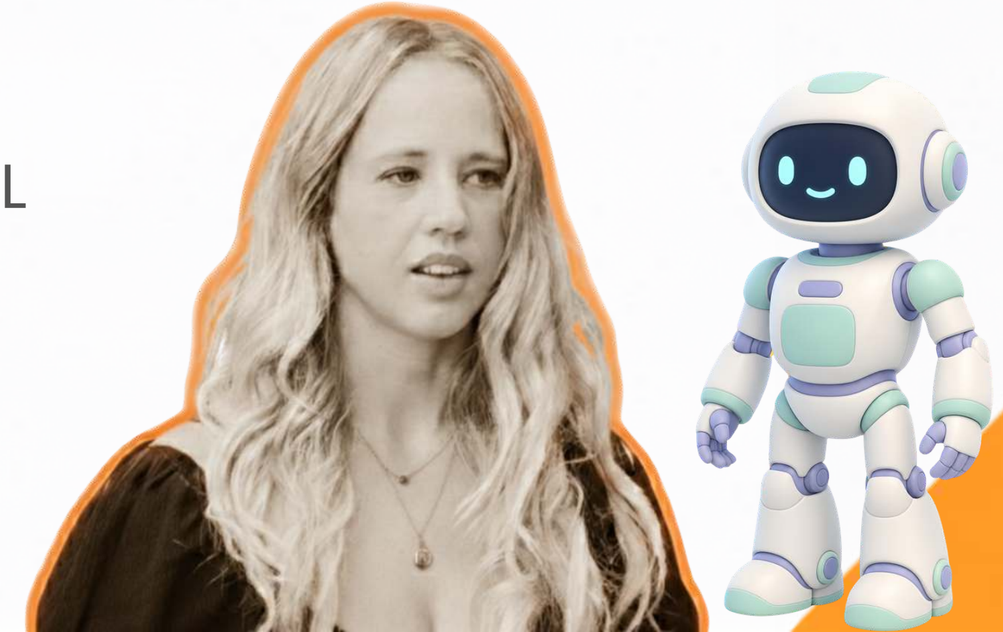
The CONTENT WOLVES

The costs of using AI for content

AND HOW TO REDUCE YOUR ENVIRONMENTAL IMPACT.

Kiri Nowak-Smith



Kiri

The CONTENT WOLVES

13 years in freelance copywriting.

17 years in SEO and Content marketing.

Director at The Content Wolves Ltd.



I ghostwrite and edit books.

Children’s books.

What I will cover

01 Awareness

02 Costs (impact)

03 Solutions

Awareness

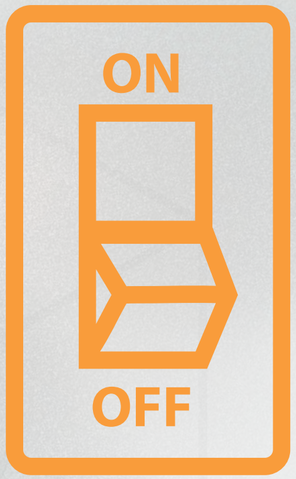
- Environmental impact.
- Education.

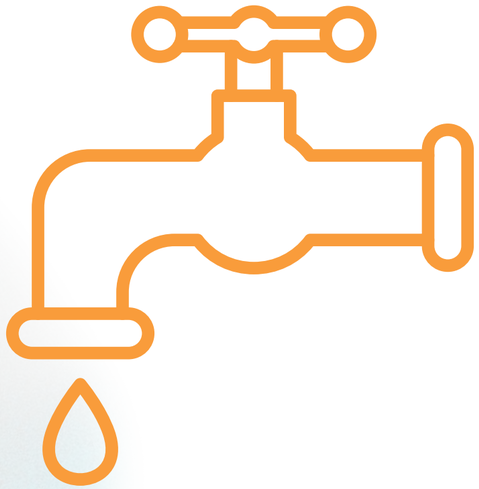




Awareness

We think about our energy use at home, so why not when we create content?





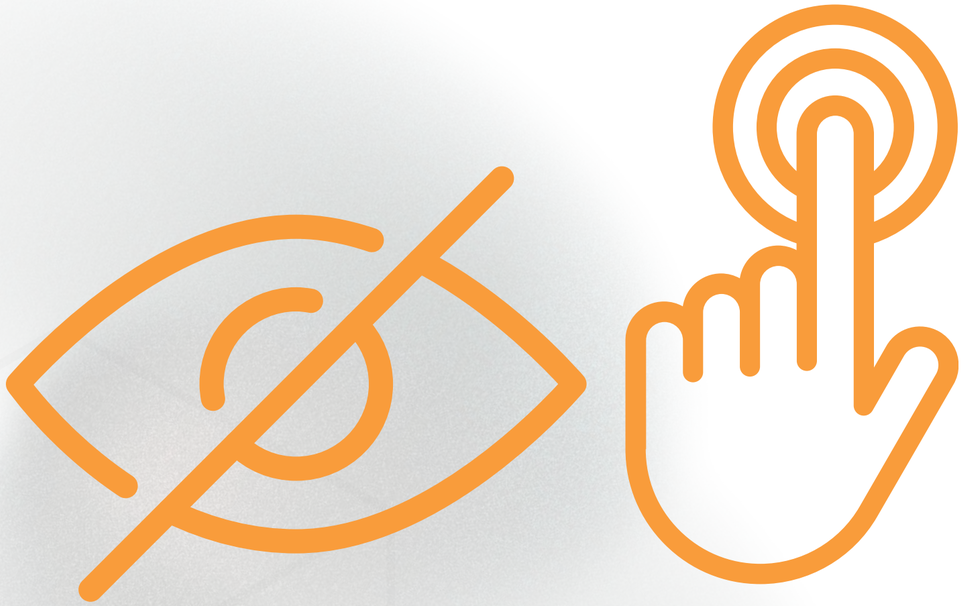


Awareness

All content has an environmental cost.



Awareness



HARD TO NOTICE

&



NO PENALTY

Other than token costs

Awareness



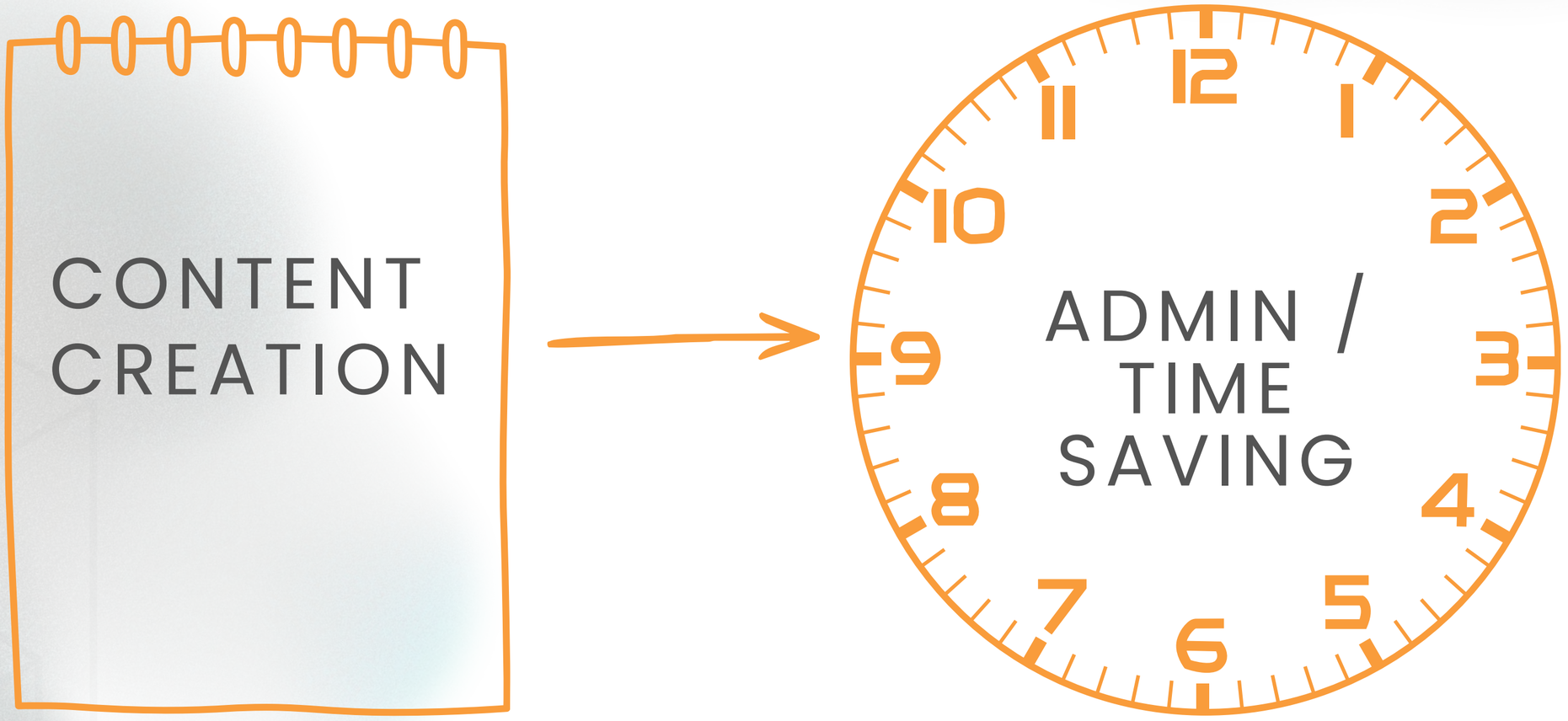
Congestion charge.



Plastic packaging tax.

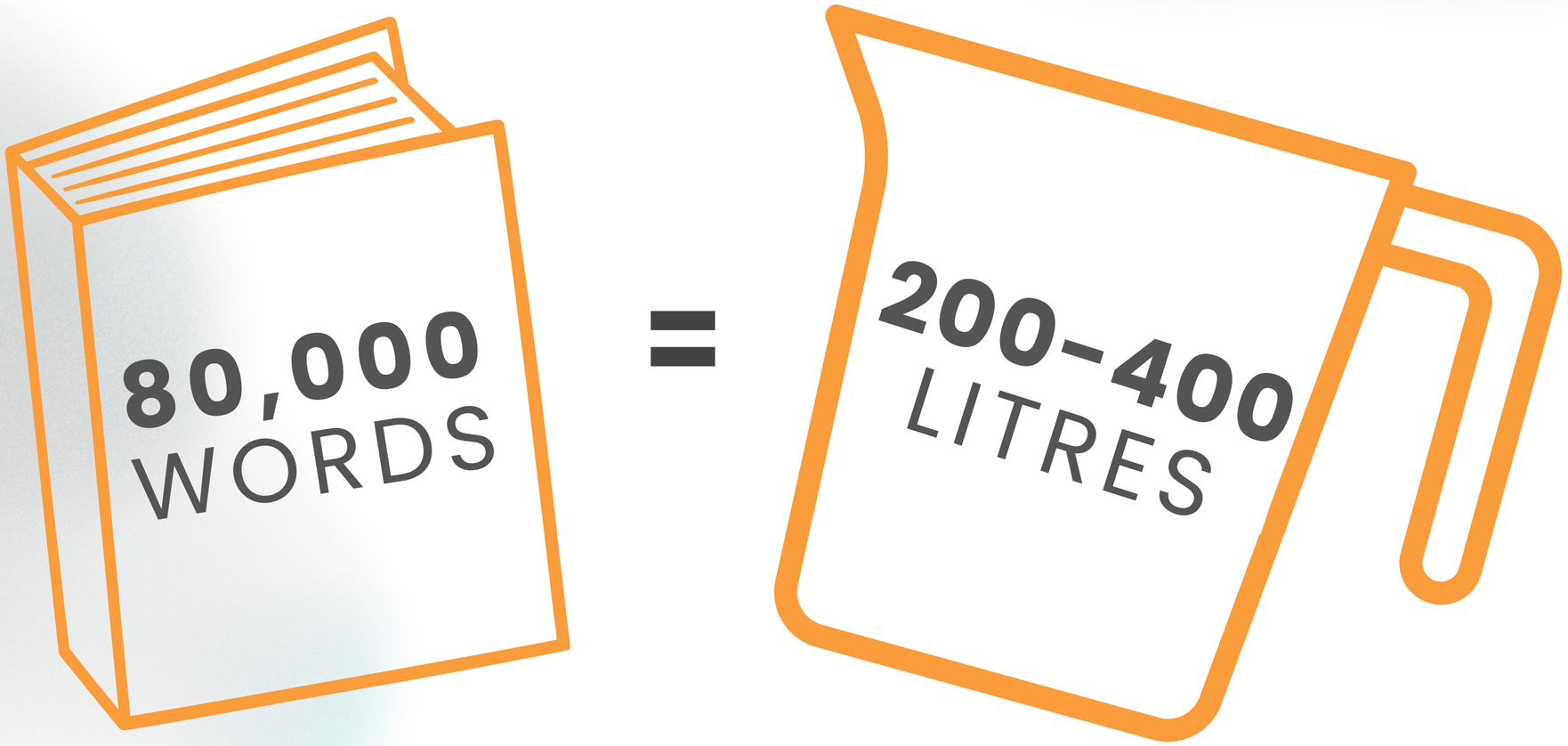
Fines for littering.

Awareness



Costs.

AI content costs



AI content costs

- I’ve worked with clients who have done this.
- More editing time.
- More energy for both AI + human.
- Not another AI book...
- Like reading a letter from a slightly eccentric distant aunt or uncle who speaks like they’re from the 1800's...

AI content costs



vs.



AI content costs

Typing a prompt to write text releases about 0.5 to 1.5 grams of carbon dioxide for advanced models like GPT-4.

The amount will depend on the size of the answer.

AI content costs

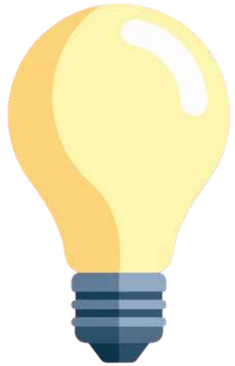


500 words.

= 10 queries.

= 3 watt-hours of electricity.

The same amount of energy an LED light bulb consumes in an hour.



AI content costs



Human content consumes energy too.

1 hour of writing = 50 watt-hours (0.05 kWh).

Charging an iPhone 18 Pro (16 watt-hours).

So can charge it roughly 3 times.

AI content costs

According to research...

Smaller models can match the environmental impact of human programmers when they succeed, though they often fail.

Standard, widely used models are far more environmentally strenuous.

AI content costs

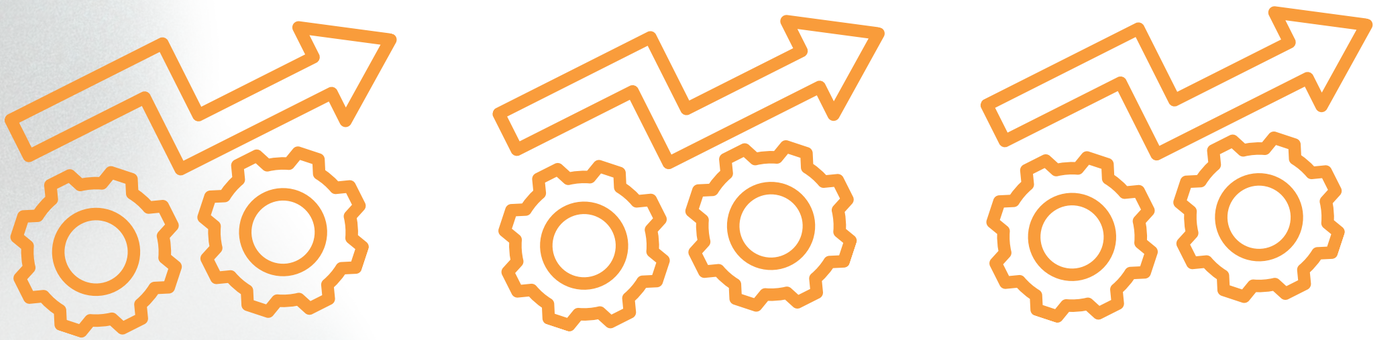


GPT-4:

= **5-19 x more** Co2 than humans.

AI content costs

AI-driven work generally consumes more upfront energy, water, and resources compared to human labour but scales far more efficiently, making it cost-effective for large tasks.



... BUT an experiment was conducted where a model failed to produce correct output even after **100** rounds of trials.

AI content costs

However, using AI to create content at scale is not recommended.

Not great for users or search engines.

Google issues penalties and ranking drops for "scaled content abuse"—mass-producing unoriginal, low-value AI content.

AI content costs

- If you generate an AI post and then spend an hour editing it, the energy consumption goes up.
- If you generate content, then don’t use it = wasteful.
- If you use it to create images and graphics = more.
- Human writers can produce a better initial output.

AI content solutions

How can you do better?

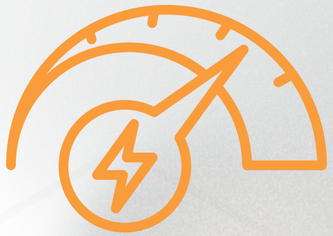
AI content solutions

- Use AI to speed up the process, not do all the work.
- If creating human content, use AI to speed up your work = less time on laptop = less energy.
- Task: 5 hours down to 2.
- Can use your computer —> more important things.
- Do double the amount of work.
- Example: I used AI to find full stops followed by two spaces.

AI content solutions

- Digital convenience comes at a cost that people don’t see.
- We need to start taking responsibility for our impact.
- Here’s how you can reduce yours...

AI content solutions



1. Monitor your usage & impact



2. Use light or local models

e.g. Gemini Flash, GPT-4o mini, Ollama.



3. Think before you generate

Don’t use blindly.
CHANGE HABITS.
Are images/graphics really needed?
Don’t use as placeholder.



4. Use reputable sources to educate

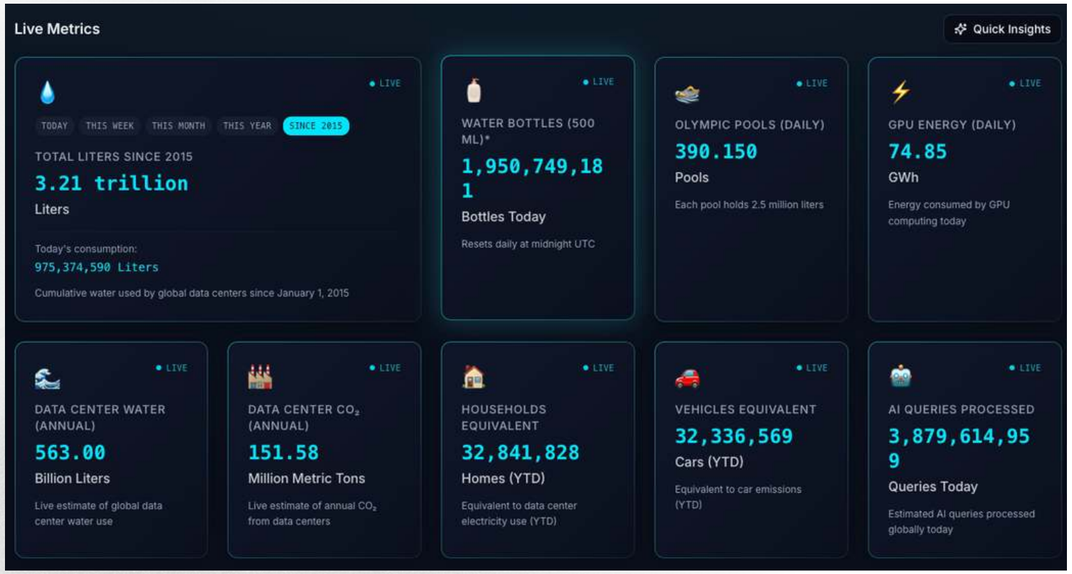
e.g. peer-reviewed academia, scientific anlayses, official organisation reports.



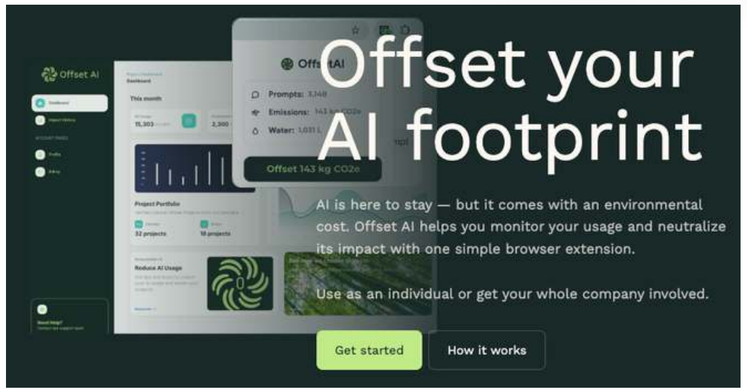
5. Use AI to improve writing

AI can train your team to write better.
=better human writers.
=less reliance on AI to generate content.

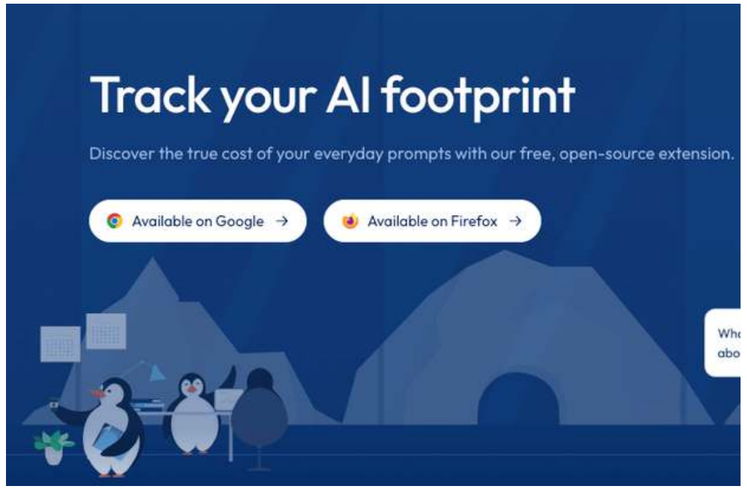
AI content solutions



Live dashboard on **watercost.live**
Check in every now and again.



offsetAI.app
Fund trusted climate and water projects.



antarctica.io
/ai-wattch

AI content solutions

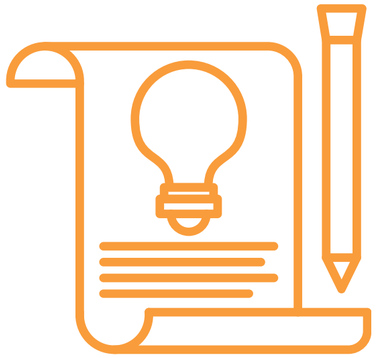


6: Spell check choices

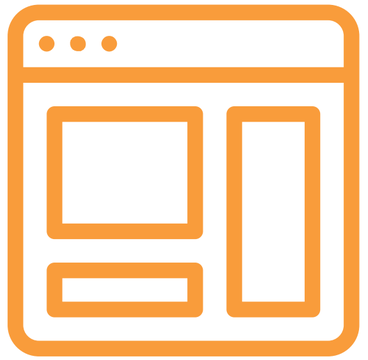
e.g. Grammarly



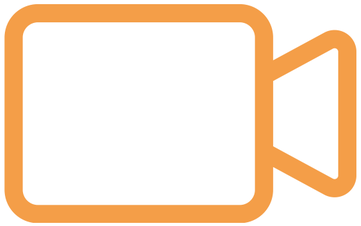
7: Batch requests



8: Provide detailed briefs & better prompts



9: Reuse templates



10: Consider content type

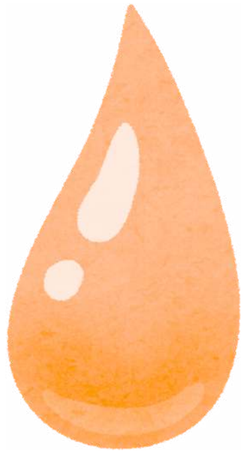
AI content solutions



One AI image two tablespoons (29 mL) .

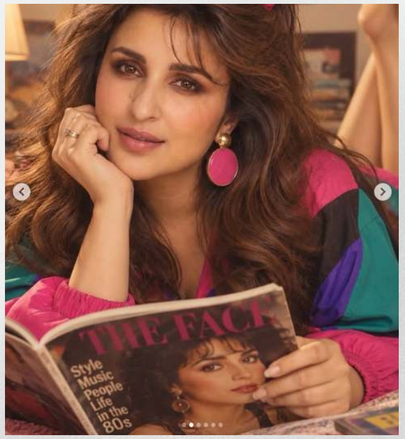


AI video generation is even more: Roughly **4.1 litres**.



Same as 1 person’s water needs for 2 days.

AI content solutions







0.1 to 4 watt-hours per image to follow trends = enough electricity to power a 10W LED bulb for about 17 minutes.

Not much for one person...

But in a 24-hour window, **2.3 million** people shared their AI-generated 80s look.

Is it worth it?

AI content solutions

Spell-checking choices.

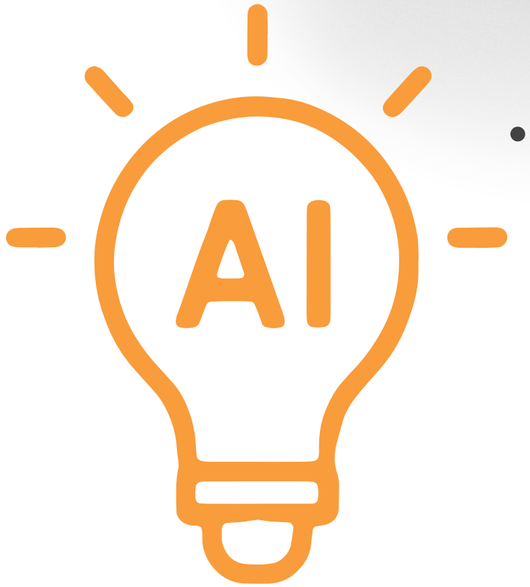
Grammarly - Relies on smaller, traditional natural language processing models rather than massive Generative AI large language models (LLMs).

Generative AI queries require up to 1,000 times more energy than traditional, localised AI classification workloads like standard spell check.

AI content solutions

Batching content requests can:

- Reduce the energy and water footprint per piece of AI-generated content.
- Makes the underlying computing more efficient.



Why does batching content work?

Let’s ask AI...

If every request requires the model to process the same background information, batching can avoid repeatedly supplying and processing that context.

- Plan first.
- Batch into fewer, well-structured requests.

AI content solutions



Example:

You want to do a range of content on solar panels.
Don’t request individual pieces of content.

“Using the following information about solar panels, create 1 blog post, 5 social posts, and 10 headlines. Keep the messaging consistent and don't repeat the background information.”

AI content solutions

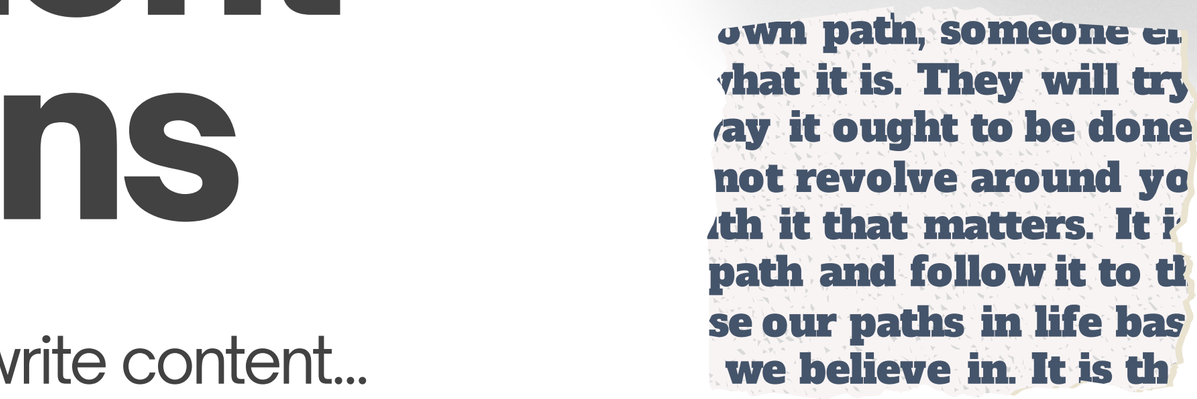
- Only use what you need (the smallest amount to get the job done).

- Better way?



e.g. graphics vs existing image libraries.

AI content solutions

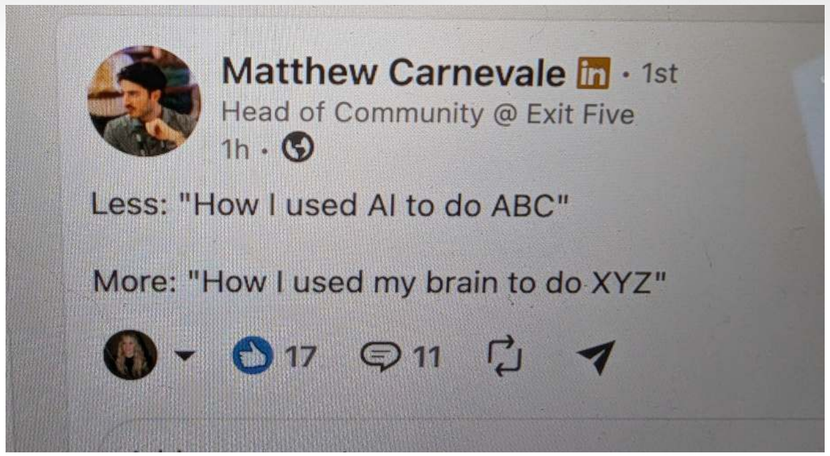


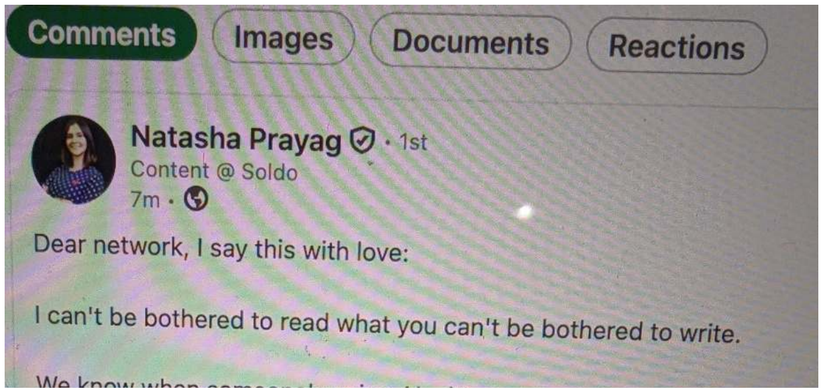
- If you must use AI to write content...
- Specify the word count.
- Give it examples. Your own content you like, plus brand guidelines = Output better/less prompts.
- Prompt better! Imagine a text vs WhatsApp message.



AI content solutions

- Using our brain for ideas? Free.
- Protect the environment, but also creativity.
- Our ability to write and communicate is not something we should hand over to machines.
- But we can use AI to speed things up and sharpen our skills.





AI content solutions

- Thank you for listening.
- Connect with me.

www.thecontentwolves.com

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https://www.instagram.com/kirinowaksmith

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